# D. J. Rasmussen, Ph.D.

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## RECENT PROFESSIONAL EXPERIENCE

**2025–now** Founder and Principal, Degree Day

**2022–2025** Lead Consultant, later Assistant Vice President (AVP), WSP (Sept. 2022–Jan. 2025)

**2022** Risk and Resilience Engineer, Arup (Jan.–Aug. 2022)

**2021–2022** Postdoctoral Research Associate, School of Public and International Affairs, Princeton University

## EDUCATION

**2021** Ph.D. in Public and International Affairs, Princeton University

Dissertation: *Coastal Defense in an Era of Sea-Level Rise: Science, Politics, and Decision-Making* [pdf]

Committee: *Michael Oppenheimer, Robert E. Kopp, Ning Lin, Guy Nordenson*

**2013** M.S. in Civil and Environmental Engineering, University of California, Davis

Thesis: *The Ozone Climate Penalty: Past, Present, and Future* [pdf]

**2009** B.S. in Atmospheric and Oceanic Science, University of Wisconsin, Madison

Thesis: *An Analysis of Climate Change Impacts on Future Wind Energy Production in California* [pdf]

## OTHER PROFESSIONAL EXPERIENCE

**2013–2016** Contracted Climate Risk Analyst, Rhodium Group

**2014–2015** Senior Engineer, ENVIRON/ Ramboll

**2012–2014** Research Assistant and then Engineer, California Air Resources Board

**2010–2011** Research Analyst, Geophysical Fluid Dynamics Laboratory/ NOAA

**2008–2010** Research Assistant, UW-Madison

## JOURNAL ARTICLES

**2026** **D.J. Rasmussen**. Multivariate bias correction of ERA5 using in-situ observations for planning and engineering. *Environmental Research: Climate*. **5** (2) 025026 doi:10.1088/2752-5295/ae63ee [pdf]

**2025** J. Morim, **D.J. Rasmussen**, T. Wahl, F.M. Calafat, R.E. Kopp, and M. Oppenheimer. US-CoastEX: Observation-based probabilistic reanalysis of storm surge and sea level extremes for the United States. *Scientific Data* 12, 1395. doi:10.1038/s41597-025-05730-1 [pdf]

**2025** J. Morim, T. Wahl, **D.J. Rasmussen**, F.M. Calafat, S. Vitousek, S. Dangendorf, R.E. Kopp, M. Oppenheimer. Observations reveal changing coastal storms around the United States. *Nature Climate Change*. doi:10.1038/s41558-025-02315-z [pdf]

**2023** T.H.J. Hermans, V. Malagón-Santos, C.A. Katsman, R.A. Jane, **D.J. Rasmussen**, M. Haasnoot, G.G. Garner, R.E. Kopp, M. Oppenheimer, and A.B.A. Slangen. The Timing of Decreasing Coastal Flood Protection Due to Sea-Level Rise. *Nature Climate Change*. doi:10.1038/s41558-023-01616-5 [pdf]

**2023** **D.J. Rasmussen**, R.E. Kopp, and M. Oppenheimer. Coastal flood protection megaprojects in an era of sea-level rise: politically feasible strategies or Army Corps fantasies? (Editor's Choice Collection) *Journal of Water Resources Planning and Management*. **149** (2) 10.1061/(ASCE)WR.1943-5452.0001613 [pdf]

**2022** **D.J. Rasmussen**, S. Kulp, R.E. Kopp, M. Oppenheimer, and B. Strauss. Popular extreme sea level metrics can better communicate impacts (Opinion). *Climatic Change*. **170** (30) doi:10.1007/s10584-021-03288-6 [pdf]

**2021** B. Strauss, S. Kulp, A. Levermann, and **D.J. Rasmussen**. Unprecedented threats to cities from multi-century sea level rise. *Environmental Research Letters*. **16** 114015 doi:10.1088/1748-9326/ac2e6b [pdf]

**2021** C. Tebaldi, R. Ranasinghe, M. Vousdoukas, **D.J. Rasmussen**, B. Vega-Westhoff, E. Kirezci, R.E. Kopp, R. Sriver, and L. Mentaschi. Extreme Sea Levels at Different Global Warming Levels. *Nature Climate Change*. doi:10.1038/s41558-021-01127-1 [pdf]

**2021** **D.J. Rasmussen**, R.E. Kopp, M. Oppenheimer, and R. Shwom. The political complexity of coastal flood risk reduction: lessons for climate adaptation public works in the U.S. *Earth's Future*. **9** (2) doi:10.1029/2020EF001575 [pdf]

**2020** **D.J. Rasmussen**, M.K. Buchanan, R.E. Kopp, and M. Oppenheimer. A flood damage allowance framework for coastal protection with deep uncertainty in sea-level rise. *Earth's Future*. **8** (3) doi:10.1029/2019EF001340 [pdf]

**2020** T. Frederikse, M.K. Buchanan, E. Lambert, R.E. Kopp, M. Oppenheimer, **D.J. Rasmussen**, and R.S.W. van de Wal. Antarctic Ice Sheet and emission scenario controls on 21st-century extreme sea-level changes. *Nature Communications*. **11** (390) doi:10.1038/s41467-019-14049-6 [pdf]

**2018** **D.J. Rasmussen**, K. Bittermann, M.K. Buchanan, S. Kulp, B.H. Strauss, R.E. Kopp, and M. Oppenheimer. Extreme sea level implications of 1.5 °C, 2.0 °C, and 2.5 °C temperature stabilization targets in the 21st and 22nd century. *Environmental Research Letters*. **13** 034040 doi:10.1088/1748-9326/aaac87 [pdf]

**2017** S.M. Hsiang, R.E. Kopp, A.S. Jina, J. Rising, M. Delgado, S. Mohan, **D.J. Rasmussen**, R. Muir-Wood, P. Wilson, M. Oppenheimer, K. Larsen, and T. Houser. Economic damage from climate change in the United States. *Science*. **356** (6345), 1362-1369 doi:10.1126/science.aal4369 [link]

**2016** **D.J. Rasmussen**, M. Meinshausen, and R.E. Kopp. Probability-weighted ensembles of U.S. county-level climate projections for climate risk analysis. *Journal of Applied Meteorology and Climatology*. **55** (10), 2301-2322 doi:10.1175/JAMC-D-15-0302.1 [pdf]

**2014** R.E. Kopp, R. Horton, C.M. Little, J.X. Mitrovica, M. Oppenheimer, **D.J. Rasmussen**, B.H. Strauss, and C. Tebaldi. Probabilistic 21st and 22nd century sea-level rise projections at a global network of tide gauge sites. *Earth's Future*. **2** (8), 383-406 doi:10.1002/2014EF000239 [pdf]

**2013** **D.J. Rasmussen**, J. Hu, A. Mahmud, and M.J. Kleeman. The ozone climate penalty: past, present, and future. *Environmental Science & Technology*. **47** (24), 14258-14266 doi:10.1021/es403446m [link]

**2012** **D.J. Rasmussen**, A.M. Fiore, V. Naik, L.W. Horowitz, M.G. Schultz, and S.J. McGinnis. Surface ozone-temperature relationships in the eastern US: A monthly climatology for evaluating chemistry-climate models. *Atmospheric Environment*. **47**, 142-153 doi:10.1016/j.atmosenv.2011.11.021 [link]

**2011** **D.J. Rasmussen**, T. Holloway, and G.F. Nemet. Opportunities and Challenges in Assessing Climate Change Impacts on Wind Energy—A Critical Comparison of Wind Speed Projections in California. *Environmental Research Letters*. **6** 024008 doi:10.1088/1748-9326/6/2/024008 [pdf]

## BOOKS

**2015** T. Houser, R. Kopp, S. Hsiang, R. Muir-Wood, K. Larsen, **D.J. Rasmussen**, M. Delgado, A. Jina, S. Mohan, P. Wilson, M. Mastrandrea, and J. Rising: *Economic Risks of Climate Change: An American Prospectus.* New York, NY: Columbia University Press. 384 pp. ISBN: 9780231174565 [link]

## SOFTWARE & DATA PRODUCTS

**2026** **D.J. Rasmussen**. *SCOPE-ERA5: Station-Calibrated Outputs for Planning & Engineering—ERA5* [Data set]. Zenodo. doi:10.5281/zenodo.15611601 [link]

**2026** **D.J. Rasmussen**. *SCOPE-CMIP6: Station Climate-Indicator Distributions by Global Warming Level* (v0.1) [Data set]. Zenodo. doi:10.5281/zenodo.21361665 [link]

**2026** **D.J. Rasmussen**. *SCOPE-CORDEX: Station Climate-Indicator Distributions by Global Warming Level* (v0.1) [Data set]. Zenodo. doi:10.5281/zenodo.21361663 [link]

**2026** **D.J. Rasmussen**. *CAHDC: Climate-Adjusted HVAC Design Conditions* (v0.1) [Data set]. Zenodo. doi:10.5281/zenodo.21330963 [link]

**2026** **D.J. Rasmussen**. *GloIDF: A Global Present-Climate Sub-Daily Rainfall Intensity–Duration–Frequency (IDF) Dataset at 30 Arc-Second Resolution* (v0.1) [Data set]. Zenodo. doi:10.5281/zenodo.21331633 [link]

**2017** **D.J. Rasmussen** and R.E. Kopp. *Probability-Weighted Ensembles of U.S. County-Level Climate Projections*

*for Climate Impact Modeling* [Data set]. Zenodo. doi:10.5281/zenodo.582327 [link]

## RESEARCH REPORTS

**2015** U. Nopmongcol, J. Johnson, B. Brashers, **D.J. Rasmussen**, T. Shah, J. Zagunis, Z. Liu, and R. Morris: *Formation of Secondary $PM_{2.5}$ in the Capital Region Study.* Prepared for Environmental and Sustainable Resource Development, Edmonton, AB. [pdf]

**2015** C. Emery, G. Wilson, **D.J. Rasmussen**, G. Yarwood: *CAMx Speed Improvements.* Prepared for Texas Commission on Environmental Quality, Austin, TX. [pdf]

**2015** J. Johnson, G. Wilson, **D.J. Rasmussen**, G. Yarwood: *Daily Near Real-Time Ozone Modeling for Texas.* Prepared for Texas Commission on Environmental Quality, Austin, TX.

**2014** S. Kemball-Cook, T. Pavlovic, J. Johnson, L. Parker, **D.J. Rasmussen**, J. Zagunis, L. Ma, G. Yarwood: *Analysis of Wildfire Impacts on High Ozone Days in Houston, Beaumont, and Dallas-Fort Worth During 2012 and 2013.* Prepared for Texas Commission on Environmental Quality, Austin, TX.

**2014** T. Houser, R. Kopp, S. Hsiang, R. Muir-Wood, K. Larsen, **D.J. Rasmussen**, M. Delgado, A. Jina, S. Mohan, P. Wilson, M. Mastrandrea, and J. Rising: *American Climate Prospectus: Economic Risks in the United States.* Oakland, CA: Rhodium Group, 194 pp. [pdf]

**2012** **D.J. Rasmussen** and M.J. Kleeman: Acceleration of CALPUFF using MPICH-2. Final Report to the California Air Resources Board. ARB Contract 11-760. [pdf]

**2010** **D.J. Rasmussen**, L.M. Keller, and M.A. Lazzara: A 20-year assessment of the frequency and intensity of McMurdo area high wind events. Preprints, $5^{th}$ *Antarctic Meteorological Observation, Modeling, and Forecasting Workshop*

## PRESENTATIONS

**2026** ASHRAE Annual Conference, Austin, TX. "Global 1-km Climatic Design Extremes at Ungauged Locations via Machine Learning Regionalization of Extreme Value Parameters," Paper Session 20: Advanced Heat Pumps, Ventilation, and Climate Data (*Poster*, AU-26-C074)

**2023** American Geophysical Union Fall Meeting, San Francisco, CA. "Storm surge reanalysis of Hurricane Idalia using a spatial Bayesian model" (NH21D-06)

**2023** American Geophysical Union Fall Meeting, San Francisco, CA. "Applying Allowances: A non-stationary, risk-based approach for advancing facility design to sea level rise and other climate hazards" (NH44A-02)

**2021** NASA Goddard Institute for Space Studies (*Invited*)

**2021** Woodwell Climate Research Center, Virtual (*Invited*)

**2021** American Geophysical Union Natural Hazards Meeting, Virtual (*Invited*)

**2021** Association of State Floodplain Managers Conference, Virtual

**2021** UC-Irvine Departmental Seminar, Virtual (*Invited*)

**2021** Coastal State Discussion Series (Rhode Island): The Future of Coastal Megaprojects, Virtual (*Invited*) [video]

**2020** American Geophysical Union Fall Meeting, Virtual

**2017** American Geophysical Union Fall Meeting, New Orleans, LA

**2015** American Geophysical Union Fall Meeting, San Francisco, CA

**2014** American Geophysical Union Fall Meeting, San Francisco, CA

**2011** American Geophysical Union Fall Meeting, San Francisco, CA

## PUBLIC OUTREACH

**2021** The largest movable structures on Earth could save New York City from another Hurricane Sandy, but will they get built?, *Princeton Research Day* [Link]

**2020** What can climate adaptation learn from what's in Grandpa's garage? A historical tale of two flood

protection megastructures, *Highwire Earth* [Link]

**2020** *Could megastructures save New York City from another Hurricane Sandy?*, short film (independently produced) [Link]

**2020** Eight years after Hurricane Sandy New Yorkers are still waiting for flood protection, *C-PREE Blog* [Link]

## AWARDS AND GRANTS

**2020–2021** Dissertation Completion Fellowship/Post-graduate Research Associate (Princeton)

**2019** Karl F. Schlaepfer '49 and Gloria G. Schlaepfer Fund (Princeton)

## PROFESSIONAL SERVICE

- American Society of Civil Engineers, Committee on Adaptation to a Changing Climate (2025–)
- U.S. Green Buildings Council, LEED v5 Resilience Working Group (2023–2024)
- Invited panelist for "Water, Water Everywhere: Climate Change, Flooding and the Gowanus Rezone", Hosted by: *Voice of Gowanus*, 2021 [Link]
- Informal Reviewer, IPCC Special Report on the Ocean and Cryosphere in a Changing Climate, 2019
- American Meteorological Society, Energy Committee (2011–2013)
- Reviewer: *Nature Climate Change; Communications Earth & Environment; Earth's Future; Environmental Research Letters; Atmospheric Chemistry and Physics; Journal of Geophysical Research: Oceans; Climatic Change; Weather and Climate Extremes; Scientific Reports; Atmospheric Environment; Atmospheric Research; Ocean & Coastal Management; Journal of Flood Risk Management; Environmental Research Communications; Environmental Research: Climate; Journal of Waterway, Port, Coastal, and Ocean Engineering*

## TEACHING

TA, *The Environment: Science and Policy* (Princeton University, Spring 2017, 2018)

## ADDITIONAL INFORMATION

**Citizenship:** USA